(Query).

Pandas vs SQL. (where).

Merge!

Df1.

Df2.

Unstructured/ Structured

(Group by) = Merge.

Relational DB (Structured).

(Group by) ⇒ Inner

outer — left. Right. full.

"Joins" → Df3 (SQL).

Joins.

1 employee works 1 Department — cross

1:1
1:m
m:m
m:1.

Emp. → m:1. → Department

EmpNo	ename	Sal	DeptNo
1001	—	—	20
1002	—	—	30
1003	—	—	10
1004	—	—	30
1005	—	—	20
1006	—		10
1007.	—	—	50

DeptNo	DName	loc
10	ACC	HYD.
20	Sales	Chennai
30	operation	Mumbai
40	Reseau	PUNE

location — Employee.

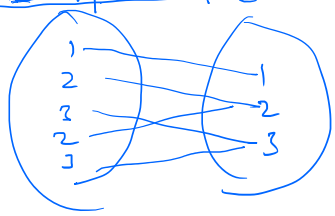
HYD — 2.

Chennai — 2.

Mumbai — 10.

A → B ∀ (A = B)

(A < B)



1001	ABCD	2500	20	Sales	Chennai
1002	XYZ	1500	30	operation	Mumbai
1003	—	—	10	Account	HYD.
NaN	NaN	NaN	40	Reseau	PUNE

Only.

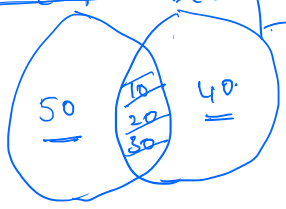
INNER (Matching)

{10, 20, 30}

Matching A ∩ B

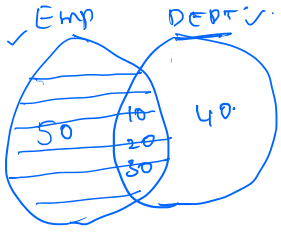
+.

Not Matching.



{10, 20, 30}.

{A}

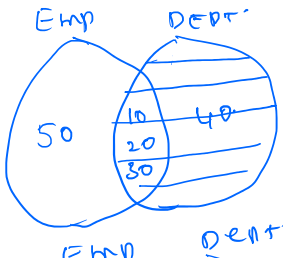


left outer Join. {10, 20, 30} + {50}.

All Data from left +

Matching from Right

{B}.



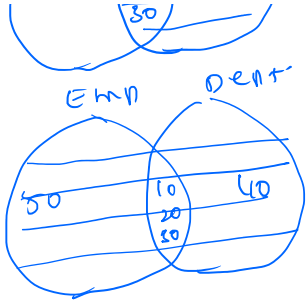
Right outer Join. {10, 20, 30} + {40}

All Data from Right +

Matching from left.

Full outer Joins. {10, 20, 30} + {40} +

(A ∪ B)



Full outer joins. {10, 20, 30} + {40} + {50}.

All Data from both Tables.

pd.merge(left,
right,
on,
how = 'inner' | left | Right | outer | cross

left_on = left_index,
Right_on = Right_index,
Validate,
Suffix,
} indicator.